# Wachirawit Kaewdang

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## SUMMARY

Computer Science graduate (Software Engineering track) from Thammasat University, working across machine learning and full-stack web development (React, Node.js). In my senior project I tested our model against a naive baseline and showed its 0.71 F1 wasn't real skill. I care about checking that results hold up, not just reporting them. Seeking Machine Learning Engineer, Software Engineer or QA roles.

## EDUCATION

**Thammasat University** 2022 – May 2026

*BSc Computer Science (Computer and Information Science major), Software Engineering track*

## PROJECTS

**Stock Price Direction Prediction – LSTM + Self-Attention with Explainable AI** Senior project (CS403)

*github.com/wachirawit-kaewdang/Project-CS403*

- Built a deep learning pipeline predicting 3-day price direction for 10 US stocks across 5 sectors (2020–2025), from 27 engineered features (momentum, volatility, moving averages, RSI, MACD, volume, S&P 500 context and bond yield).
- Architecture: LSTM with self-attention (3 layers, hidden size 128) over a 60-day window, trained with Focal Loss and class weights, and compared with Random Forest, Logistic Regression, a soft-voting ensemble and a VIX feature set.
- Compared SHAP (KernelExplainer) with Pearson correlation: SHAP ranked bond yield top-5 for 6 of 10 stocks (#1 for JPM and AMZN); correlation agreed for JPM but split for TSLA, COST and XOM (TSLA: 2nd by SHAP, 15th by correlation). Overall rank gaps of 5–12 positions were close to the 9 expected by chance, which we documented.
- Evaluated results against a naive baseline: on 7 of 10 stocks the tuned model collapsed to always predicting "up", so its F1 matched an always-up model; the ensemble's threshold-free ROC-AUC averaged 0.568, close to random. Identified threshold and early-stopping leakage and documented fixes.
- My contribution: data collection and preparation, training approach and model architecture, joint SHAP analysis, and the baseline check. Two-person project.

Python · PyTorch · SHAP · scikit-learn · pandas · NumPy · Jupyter

**This Portfolio and Its Data Checks** Personal project

- React + Vite site whose CI rebuilds its metrics and tests both pages in headless Chrome. I set the goals, chose between proposed approaches and approved them, reviewed every change, and ran regular design and code critiques; Claude Code wrote most of the code.

**QA Automation Exercise – Robot Framework** Take-home exercise

- 11 passing automated tests with reports, covering UI login flows (SeleniumLibrary), a REST API (RequestsLibrary) and a Python script, plus 18 written test cases for a mobile login flow, prioritised High to Low.

**Recipe Web Application – Full-Stack** CS369, group of 5

- React + Vite frontend with client-side routing, connected to a Node.js / Express REST API.
- My contribution: the MyRecipe management feature, recipe deletion, and frontend–backend integration.

React · Vite · JavaScript · Node.js · Express · REST APIs

**HTTP Web Server from Scratch** CS322 Computer Networks

- Implemented an HTTP/1.0 server in Python using raw socket programming – no frameworks – serving an HTML page and a GIF with the right content-type headers, and returning 404 for anything else.

Python · sockets · TCP/IP · HTTP

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, JavaScript, Java, C, C++, SQL |
| **ML & Data** | PyTorch, scikit-learn, SHAP, pandas, NumPy, Jupyter – time series, feature engineering, explainable AI |
| **Web** | React, Vite, Node.js, Express, Django, REST API design, HTML/CSS |
| **Databases** | MySQL, MongoDB |
| **Tools** | Git, Docker, automated testing (TDD), Robot Framework, object-oriented design |
| **Spoken** | Thai (native), English (conversational) |